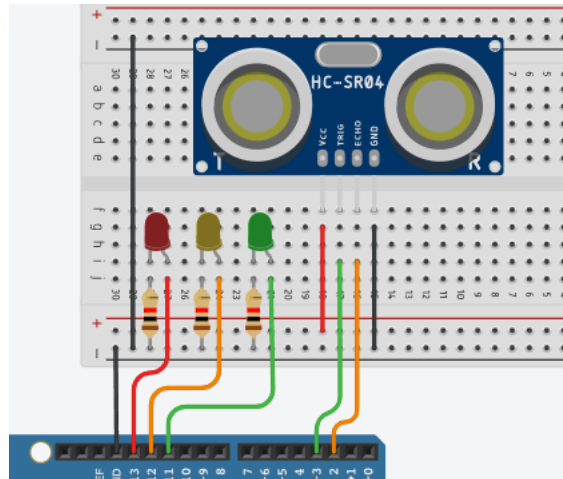


Mock Test 2: Arduino Security System with Ultrasonic Sensor

Duration: 2 Hours

Scenario:

Design and implement a security system using an Arduino UNO and an HC-SR04 ultrasonic distance sensor. The system should detect objects that come within a certain distance threshold and trigger an alarm (red LED) when the system is armed.



Requirements:

1. Use an HC-SR04 ultrasonic sensor to continuously measure distance
2. Use the `map()` function to convert the raw sensor reading to distance in millimeters
3. Implement a password system ("Arm") to arm/disarm the security system
4. Create a visual indicator system using LEDs:
 - Green LED (pin 11): Flashes when system is in standby mode
 - Yellow LED (pin 12): Stays on when system is armed
 - Red LED (pin 13): Turns on when an intruder is detected (distance threshold breached)
5. Display distance measurements via Serial monitor
6. The system should only trigger the alarm when armed and the distance threshold is breached

Instructions:

1. Connect the HC-SR04 sensor to pins 2 (ECHO) and 3 (TRIGGER)
2. Connect the three LEDs to pins 11 (green), 12 (yellow), and 13 (red) with appropriate resistors
3. Write code that uses the `map()` function to convert the sensor's raw pulse duration values to distance in millimeters
4. Implement the password system to arm/disarm via Serial input

5. Follow the provided circuit diagram and sample code structure to complete your implementation

Expected Outcomes:

1. When running, the system should continuously display distance readings in millimeters
2. The green LED should flash in standby mode
3. When "Arm" is entered via Serial monitor, the system should arm and the yellow LED should light up
4. When armed, if an object comes closer than the threshold distance, the red LED should light up

Requirement	Complete?
HC-SR04 pins correctly defined (2: ECHO, 3: TRIGGER)	☐
LED pins correctly defined (11: green, 12: yellow, 13: red)	☐
<code>map()</code> function used to convert to millimeters	☐
Password variable set to "Arm"	☐
Serial communication initialized (9600 baud)	☐
Distance continuously measured and displayed in mm	☐
Green LED flashes in standby mode (500ms on/off)	☐
Yellow LED turns ON when password "Arm" entered	☐
Red LED turns ON when distance < threshold (while armed)	☐
System maintains armed state until reset	☐
Distance readings change when objects move	☐
Password entry activates armed mode	☐
Moving object close triggers red LED when armed	☐
System completes multiple test cycles successfully	☐